

Phase 3



1. Software Architecture

System Architecture

The KeyShare Platform System is a car rental booking application. This application allows people to rent cars, car owners to list their vehicles, business users to manage vehicles and administrators to control everything.

The KeyShare Platform System follows a three-tier architecture.

Frontend:

This is the part of the system that people see. Renters, car owners, business users and administrators all use the frontend. On this part of the system people can sign up log in search for cars look at car details, book cars manage their listings send messages leave reviews and get notifications.

Backend:

This is the part of the system that does all the work. It takes care of things like logging in listing cars searching for cars booking requests and checking if a car is available. It also handles reviews, messages, notifications and things that administrators do. Sometimes it even handles payments.

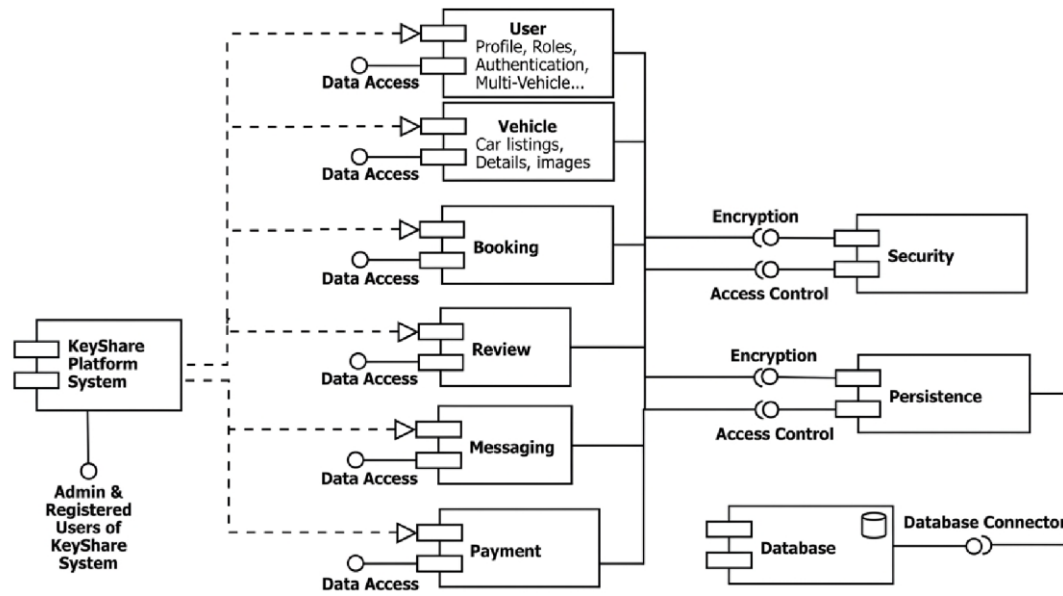
Database:

This is where all the important information is stored. It has information about people, cars, bookings, reviews, messages, notifications and payments. It also helps make sure that someone does not book a car that is already taken.

Example System Flow:

1. A person who wants to rent a car logs in to the KeyShare platform.
2. The person searches for a car by location or price or type of car.
3. The frontend sends this search to the backend.
4. The backend looks in the database for cars that match what the person is looking for. That are available.
5. The database sends the cars back to the backend.
6. The backend sends the results back to the frontend.
7. The person picks a car. Confirms that they want to book it.
8. The backend checks the database again to make sure the car is not already booked. 9. If the car is available the booking is saved in the database and the status is Pending.
10. The car owner gets a notification.
11. The owner says yes or no to the booking.
12. The backend updates the booking status, in the database.
13. The person who rented the car gets a notification that tells them if the booking was successful or not.

Component diagram (UML)



Component Diagram of KeyShare Platform System

Components

User component: this is where the system keeps track of user information. This includes things like profile details and roles. It also handles authentication and account-related actions. The system supports types of users such as renters, vehicle owners, business users and admins.

Vehicle component: this part of the system manages vehicle listings. It includes things like car details, price, location and availability information. Vehicle owners can use this component to add, edit or remove vehicle listings. They can also upload images of their vehicles.

Booking component: this is where the system processes booking requests from renters. It handles things like creating bookings approving or rejecting bookings and canceling bookings. The system also tracks the status of bookings. Makes sure that the same vehicle is not booked twice.

Review component: this part of the system manages ratings and comments from users. It helps to display feedback and reliability information for vehicles and users. This way users can see what other people think of a vehicle or a user.

Messaging component: this component handles communication between renters and vehicle owners. They can send each messages through the system. The system keeps track of these conversations.

Payment component: this part of the system handles payments for bookings. It stores information like the amount paid the payment status and whether the payment was successful or not.

Data Access component: this is how each component can read from and write to the database. It helps to retrieve, save and update information related to users, vehicles, bookings, reviews, messages and payments.

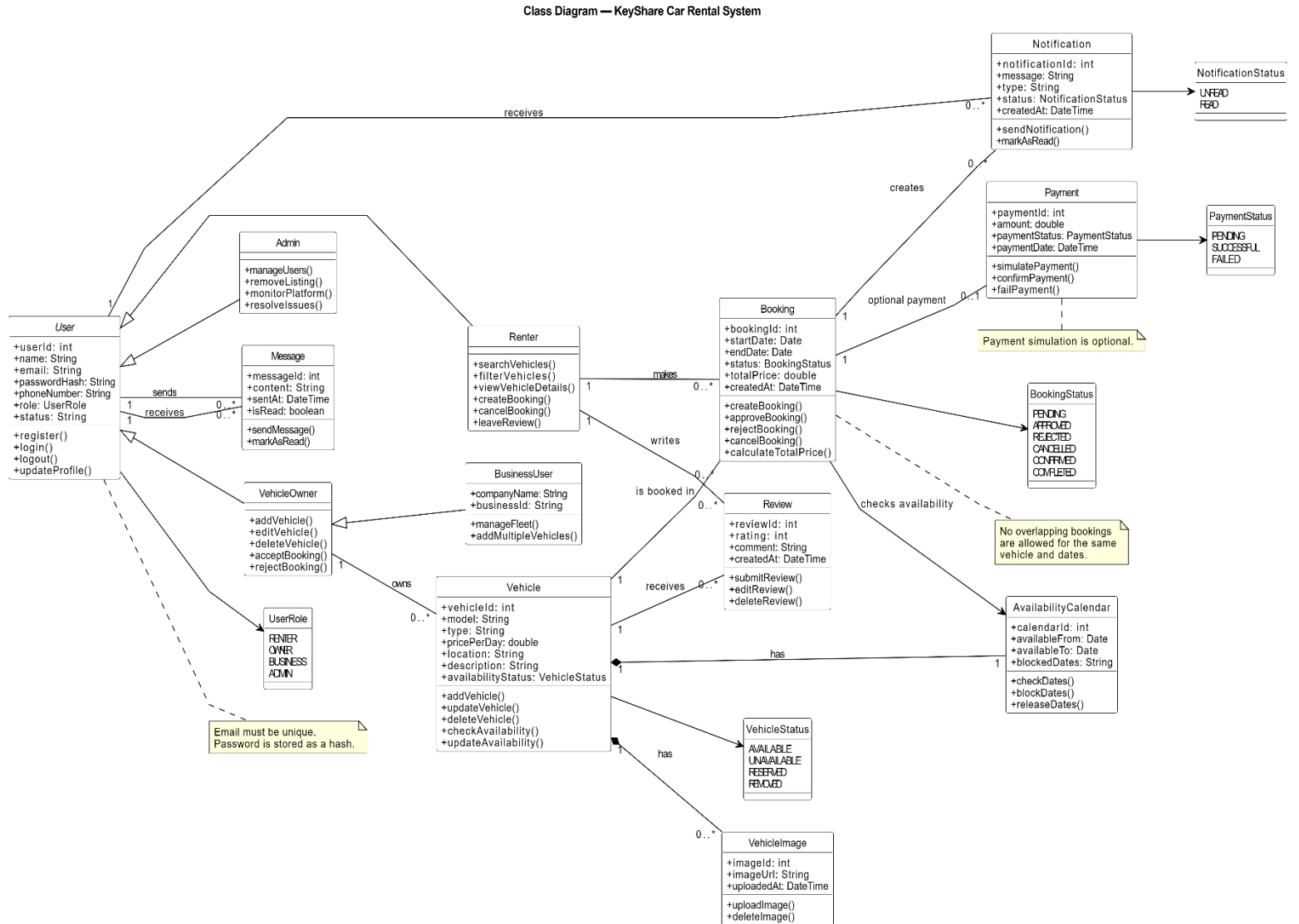
On the left side of the diagram there is the KeyShare Platform System. This is the system that connects users with the internal platform components. There are also actors who interact with the platform, such as renters, vehicle owners, business users and admins. These are the people who use the KeyShare Platform System.

On the right side of the diagram there are a few more components.

- * Security: this handles authentication, encryption and access control. It makes sure that only authorized users can access functions.
- * Persistence: this manages the storage and retrieval of application data. It ensures that the application components and the database are consistent.
- * Database: this stores the data of the platform. This includes things like users, vehicles, bookings, reviews, messages and payments.
- * Database Connector: this provides the connection, between the persistence layer and the database. It helps the system to store and retrieve data from the database.

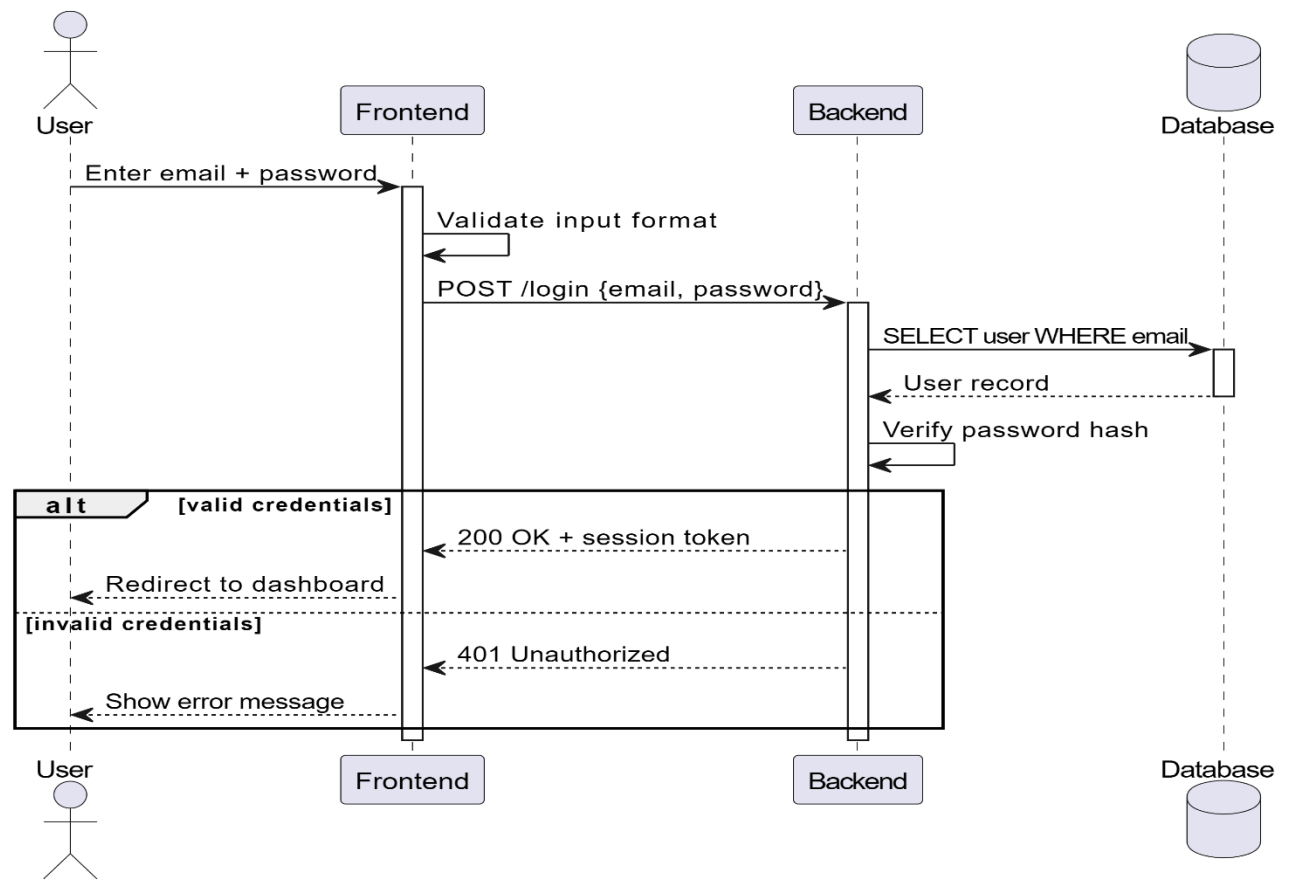
2. Detailed Design

Class Diagram

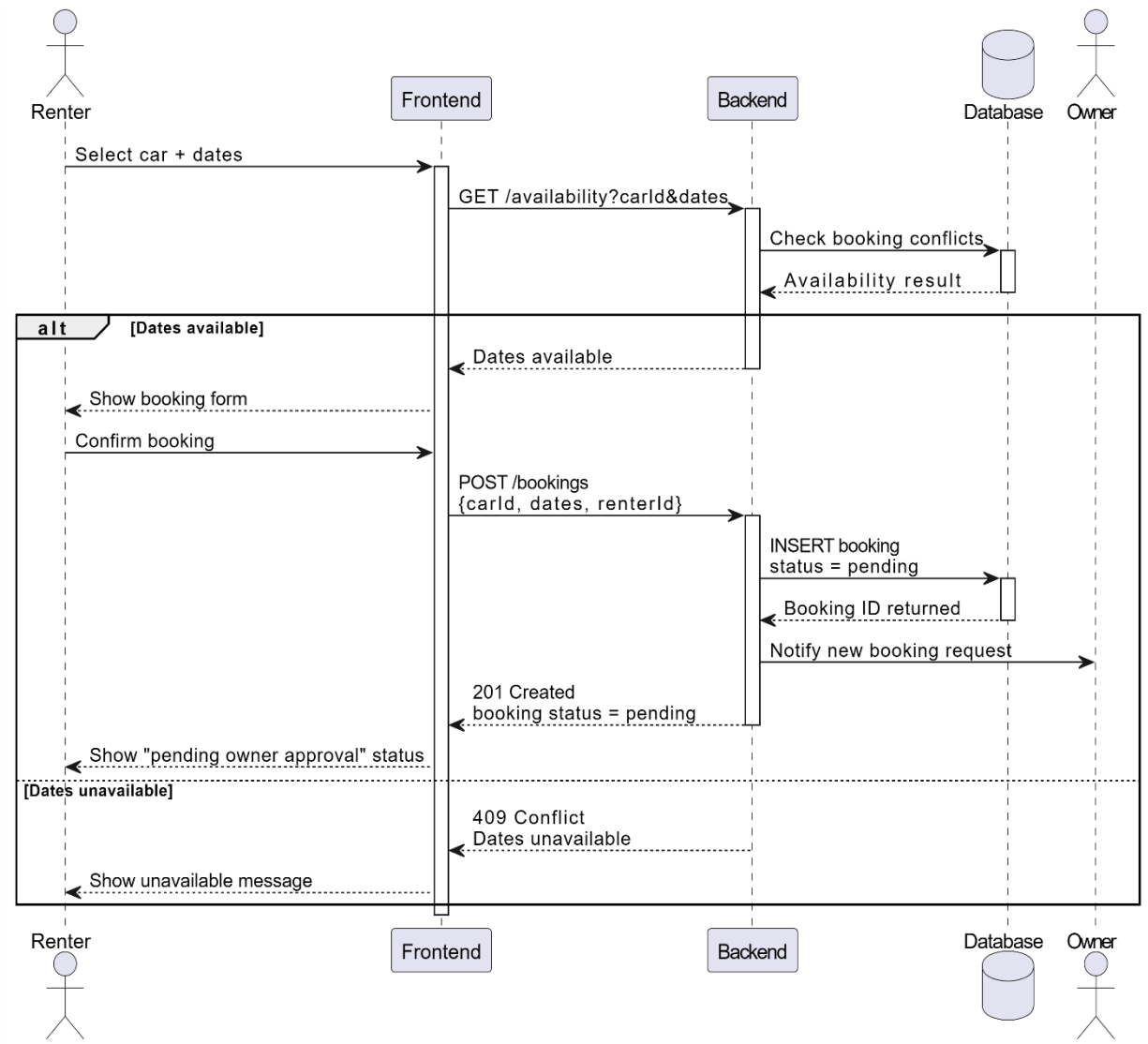


Sequence Diagram

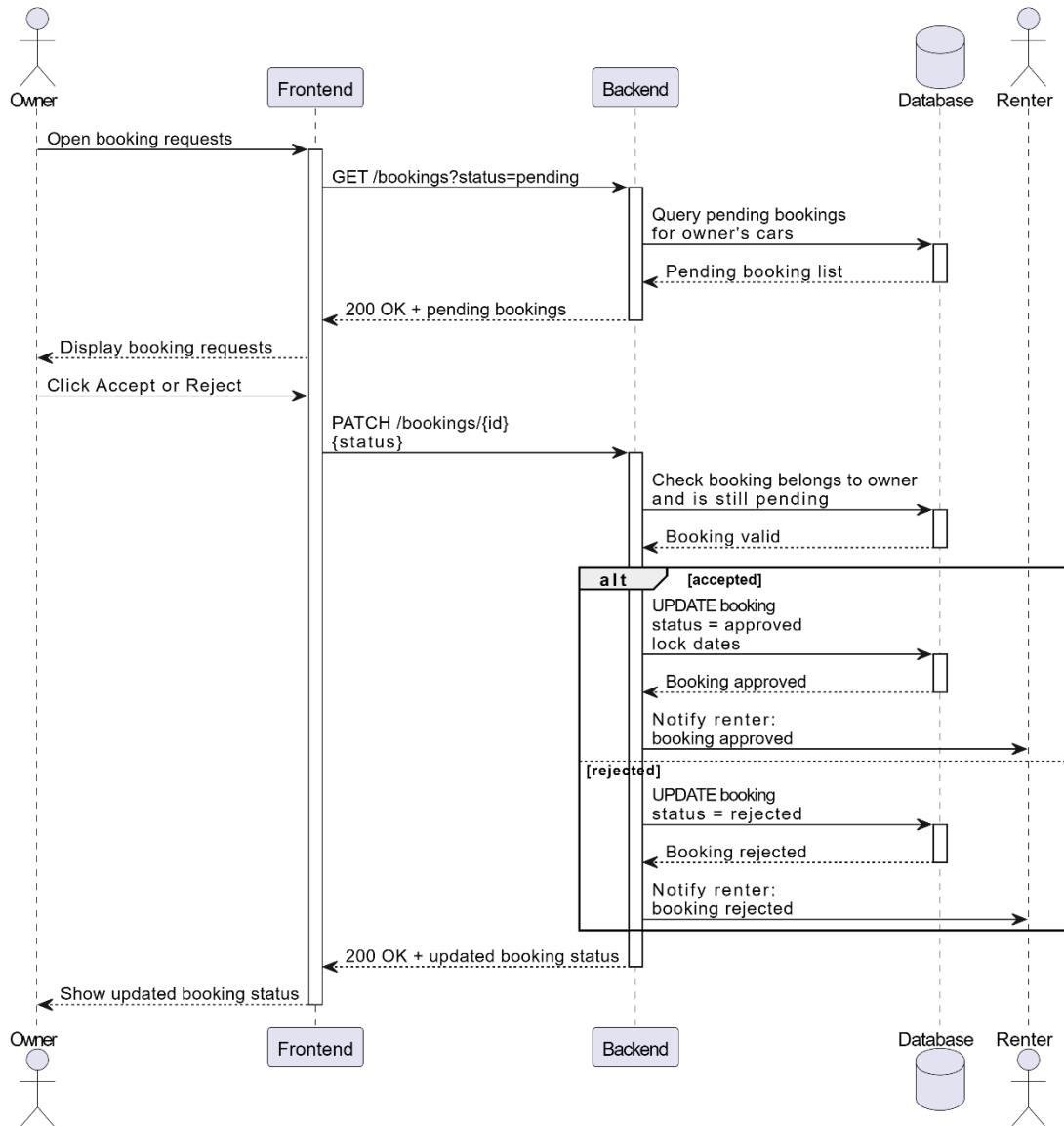
1. User login

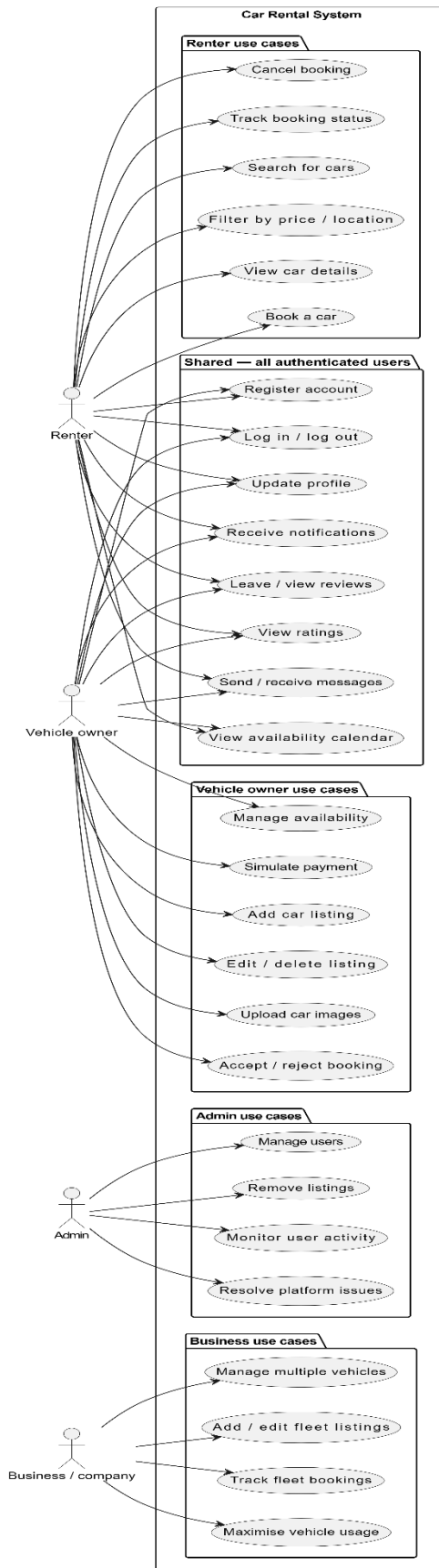


2. Renter books a car



3. Owner accepts or rejects a booking

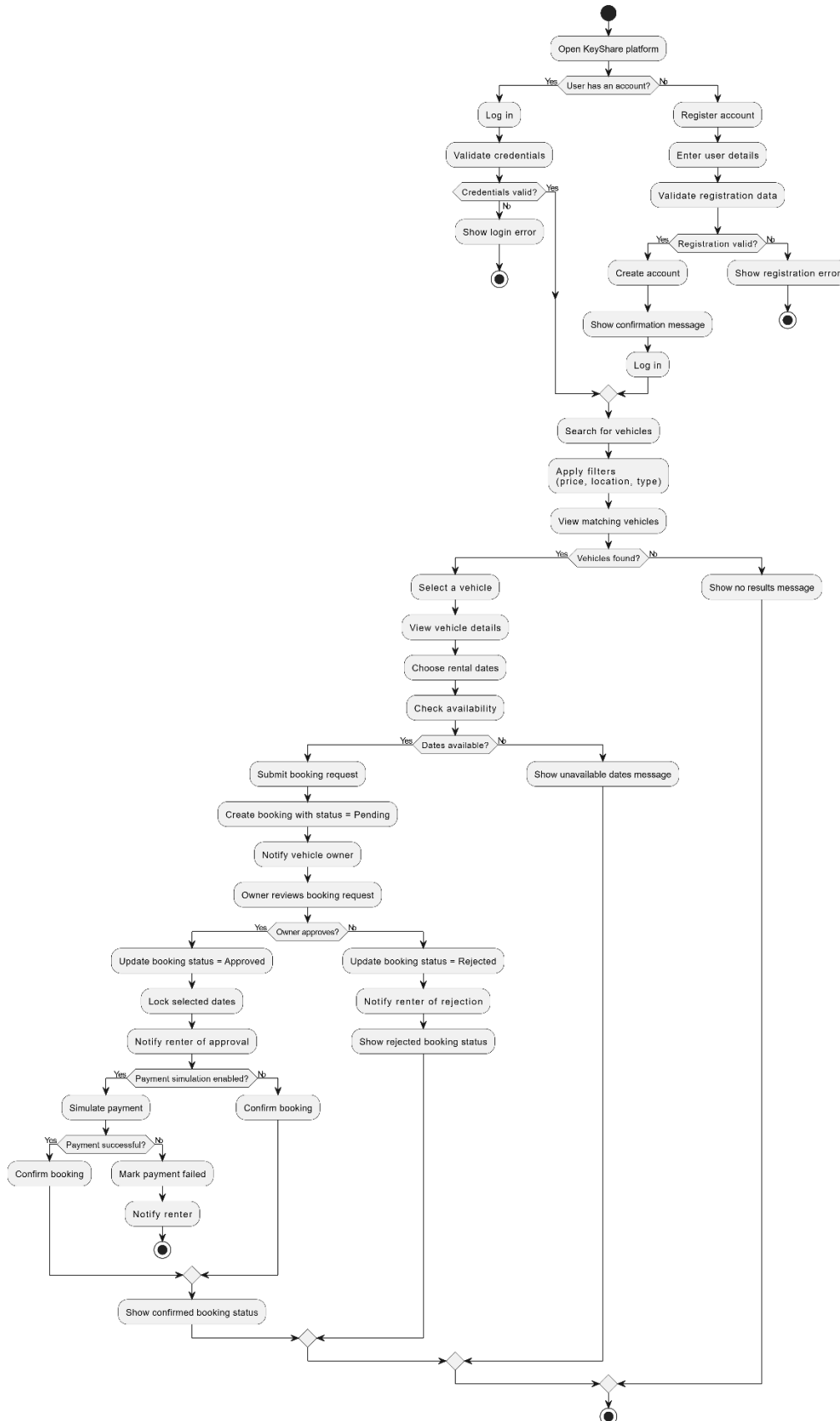




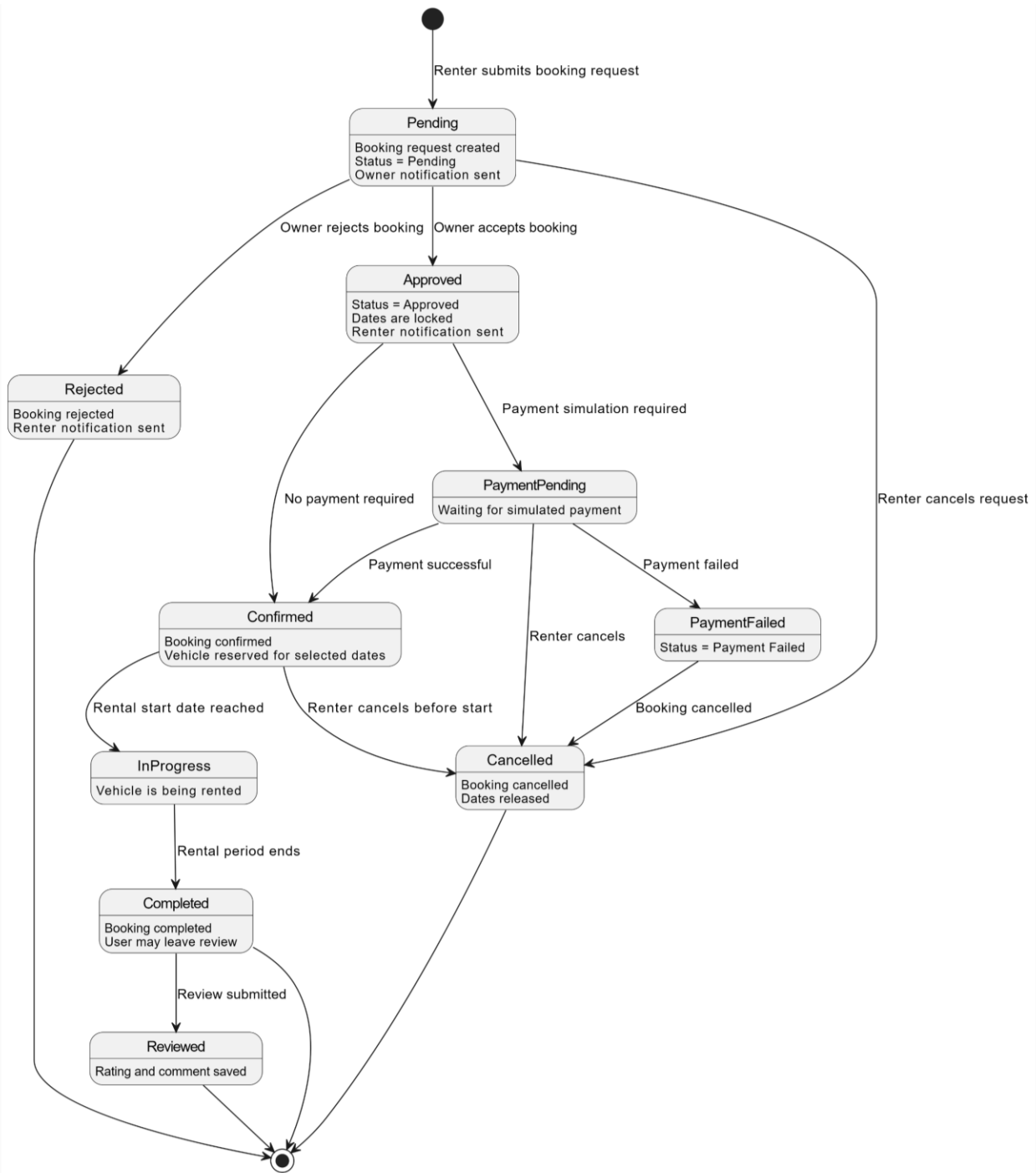
3. Modeling

Use case

Activity Diagram (Keyshare vehicle booking process) State Diagram



1. Booking status in KeyShare



2. Vehicle Listing Status in keyShare

